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1. Title of the invention:

UAV-Based CNN System for Detection and Geolocation of Underground Pipeline Deformation

2. Field /Area of invention:

- UAV-based monitoring of underground industrial pipelines (Aligned with SDG 9: Industry, Innovation and Infrastructure)
- AI/ML-driven detection of pipeline deformation using CNN and real-time vision models (Aligned with SDG 9: Industry, Innovation and Infrastructure)
- Thermal imaging for subsurface anomaly and leak identification. (Aligned with SDG 12: Responsible Consumption and Production)
- GPS-based geolocation and spatial mapping of detected defects. (Aligned with SDG 11: Sustainable Cities and Communities)
- Edge-computing-enabled real-time inspection and predictive maintenance. (Aligned with SDG 8: Decent Work and Economic Growth)
- Safety and environmental risk reduction in oil, gas, and water pipelines**.** (Aligned with SDG 6 – Clean Water and Sanitation; SDG 13 – Climate Action)

3. Prior Patents and Publications from literature (provide a table summarizing the prior art)

Sl. No.	Patent / Publication Title	Patent No. / Source	Country	Year	Key Features	Limitation / How Your Project Is Unique
1	Underground infrastructure sensing using unmanned aerial vehicle (UAV)	US11719853B2	USA	2023	Utilizes UAV with various sensors (e.g., GPR, visual) to collect data for underground infrastructure such as pipes.	The existing methods only identify pipelines or surface indicators, whereas the proposed system uses CNN-based deep learning to detect, classify, and geolocate pipeline deformations.
2	A fully intelligent inspection system for oil and gas pipelines based on UAV remote sensing and deep learning	CN210511074U	China	2020	A UAV captures pipeline images, detects safety anomalies in real time using deep learning, and transmits location-based alerts.	The system is working for surface-only threat detection. But the proposed system enables underground deformation detection using thermal/ML fusion with precise subsurface geolocation.

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3	Detection of buried pipelines and spills	US12228695B2	USA	2025	Uses aerial (drone) and geophysical data combined with *machine-learning models* to identify buried pipeline locations and possible spills. Outputs result on a geographical map.	Unlike pipeline or spill localization methods, the proposed system detects and classifies structural deformation anomalies.
4	A kind of underground utilities detection recognition method based on deep learning.	CN109685011A	China	2019	Uses CNN-based deep learning to detect underground pipelines and reduce human interpretation errors.	Unlike pipeline-only recognition, the proposed system integrates UAV-based thermal sensing with CNNs for deformation detection and analysis.
5	A kind of gas pipeline leakage detecting system based on unmanned plane	CN105805560B	China	2017	UAV detects pipeline gas leaks and diffusion using lasers and surface temperature anomalies.	Unlike gas-only detection, the proposed system identifies structural deformations using CNN, thermal imaging, and geolocation.
6	Underground pipeline detection method and system based on ground penetrating radar and deep learning	CN112130132A	China	2023	Uses GPR and YOLOv3 to detect pipelines and geolocate them with RTK accuracy.	Unlike ground-only GPR, the suggested system uses UAVs with thermal/visual ML to detect and geolocate deformations.

References:

[1] Underground infrastructure sensing using unmanned aerial vehicle (UAV), US11719853B2, 2023.

Link: https://patents.google.com/patent/US11719853B2/en

[2] A fully intelligent inspection system for oil and gas pipelines based on UAV remote sensing and deep learning, CN210511074U, 2020.

Link: https://patents.google.com/patent/CN210511074U/zh

[3] Detection of buried pipelines and spills**,** US12228695B2, 2025.

Link: https://patents.google.com/patent/US12228695B2/en

[4] A kind of underground utilities detection recognition method based on deep learning, CN109685011A, 2019.

Link: https://patents.google.com/patent/CN109685011A/en

[5] A kind of gas pipeline leakage detecting system based on unmanned plane, CN105805560B, 2017.

Link: https://patents.google.com/patent/CN105805560B/en

[6] Underground pipeline detection method and system based on ground penetrating radar and deep learning, CN112130132A, 2023.

Link: https://patents.google.com/patent/CN112130132A/en

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4. Summary and background of the invention (Address the gap / Novelty)

Industrial pipeline networks (for oil, gas, water, *etc.*) are essential but vulnerable assets. Over time, they can suffer from underground deformations caused by corrosion, soil shifting, mechanical stress, and environmental factors. Traditional inspection techniques such as manual surveys, ground-based sensing, or basic UAV imaging face challenges including limited subsurface visibility, high operational costs, slow detection rates, and poor real-time diagnosis. Recent technological advances include UAV inspections, thermal imaging, and isolated AI/ML models for surface anomaly detection. However, these approaches often fail to integrate all critical components necessary for identifying, classifying, and precisely mapping underground pipeline deformations**.** They lack holistic real-time processing, deep learning-based classification, multimodal data fusion, and geolocation accuracy -all of which are key to early detection and predictive maintenance.

The present invention bridges these gaps by introducing an integrated framework that utilizes UAVs equipped with thermal cameras to acquire high-resolution aerial data and applies CNN- and YOLO v8s-based models for automated detection and classification of pipeline deformations. Edge computing enables real-time onboard processing, while multimodal fusion of thermal and visual imagery provides richer anomaly features and improved detection accuracy. Precise GPS-based geolocation maps defect locations on the ground, and instant alerts with visual reports are transmitted to maintenance teams for proactive intervention. By unifying sensing, artificial intelligence, geospatial mapping, and automated alerting, this comprehensive system significantly enhances the accuracy, speed, and coverage of underground pipeline monitoring.

Novelty of the invention –

Missing Features in the existing Pipeline deformation and Location Systems:

1. **Lack of Integrated Multimodal Sensing**
   - They do not fuse thermal radiometric data with visual imagery*,* which limits their ability to detect subtle underground deformation signals that may not be visually apparent or thermally distinct in isolation.
2. **Absence of Real-Time AI-Driven Classification**
   - There is no use of advanced CNN architectures (e.g., YOLO / ResNet / Transformer-augmented models) for real-time detection and classification of underground deformation patterns.
3. **No Geospatial Localization of Subsurface Anomalies**
   - Precise GPS geolocation of detected deformation sites — enabling exact ground mapping and actionable intervention — is largely absent.
4. **Limited Edge Computing / Onboard Real-Time Processing**
   - Edge computing for on-the-fly analysis directly on the UAV, reducing latency, bandwidth usage, and enabling real-time alerts, is not implemented in most prior works.
5. **Gap in Predictive Maintenance and Early Anomaly Recognition**
   - There is a lack of predictive anomaly recognition — detecting hidden deformation before critical failure.

How our prototype Solves These Gaps:

1. **Integrated Multimodal Sensing**
   - Our prototype captures thermal images allowing the AI model to reason across modalities improving detection of buried structural anomalies that might be invisible in a single sensing band.

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2. **Deep Learning–Based Real-Time Classification**
    - Deploys optimized deep learning models that Detect deformation patterns in real time, classify deformation types (crack, bend, subsidence, etc.), Reduce false positives through learned feature representations.
3. **GPS Geospatial Localization**
    - Each detected anomaly is associated with geolocation metadata, mapped on a real-world coordinate system Enabling maintenance crews to locate defects rapidly and accurately.
4. **Onboard Edge Processing**
    - Processes data directly on the UAV's computing platform, which reduces dependency on ground stations, eliminates network latency, provides instant anomaly alerts.
5. **Predictive Maintenance Capability**
    - By integrating deep learning's feature extraction, thermal anomalies, temporal sequence input analysis, your system can identify subtle precursors of pipeline deformation before catastrophic failures occur, enabling predictive maintenance and not just reactive inspection.

5. Objective(s) of Invention

- To develop a UAV-based inspection system capable of monitoring large-scale industrial pipeline networks efficiently and safely without the need for manual or ground-based inspection methods.
- To detect underground pipeline deformations such as cracks, subsidence, leaks, and soil heaving by utilizing thermal and visual imaging sensors mounted on the UAV.
- To employ deep learning models, including Convolutional Neural Networks (CNNs) and real-time object detection algorithms, for accurate classification and localization of pipeline anomalies.
- To enable real-time onboard processing using edge computing devices, thereby reducing latency and eliminating dependence on remote servers or cloud-only processing.
- To integrate GPS-based geolocation for precise mapping of detected pipeline defects, allowing maintenance teams to identify exact ground locations.
- To provide real-time visualization through dashboards displaying annotated images and geospatial coordinates for rapid decision-making.
- To improve inspection accuracy, reliability, and cost-effectiveness compared to traditional pipeline monitoring and inspection techniques.

6. Working principle of the invention (in brief)

The proposed invention operates on a Raspberry Pi 5 mounted on a UAV-based multimodal inspection framework in which a drone equipped with thermal camera (MLX90640), GPS(NEO-6M), and an onboard edge computing device autonomously surveys pipeline corridors and acquires synchronized thermal and geolocation data. The captured data is preprocessed and then analyzed in real time using CNN models for deformation classification and YOLO v8s-based detection for precise localization of anomaly regions. Detected pipeline deformations are geotagged with accurate spatial coordinates and timestamps, enabling instant alerts and visualization on a monitoring dashboard. By integrating aerial sensing, deep learning inference, edge processing, and geospatial mapping into a single workflow, the system enables real-time detection, localization, and predictive maintenance of underground pipeline deformations.

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7. Description of the invention in detail (Include drawing and or photograph as needed)

A drone-mounted thermal imaging system is deployed to inspect underground pipelines for leaks and structural deformations. The onboard thermal camera captures raw temperature matrix data representing spatial temperature variations above the buried pipeline. A Plasma colour map is applied to this temperature matrix to generate a visually interpretable thermal image that highlights thermal anomalies. This colour-mapped thermal image is then transmitted to a Raspberry Pi–based edge computing unit, where image calibration, noise normalization, and pipeline-specific region-of-interest (ROI) extraction are performed to isolate the relevant inspection area.

The pre-processed thermal image is subsequently analysed using a convolutional neural network (CNN), which classifies the observed regions into normal conditions or anomalous thermal hotspots indicative of potential leakage or deformation. For precise spatial localization, a YOLO-based detection module is employed to generate bounding boxes around detected anomalies along with associated confidence scores. Each identified event is further enriched with GPS coordinates and time stamps, enabling accurate geospatial referencing. The final outputs are visualized on a monitoring dashboard that presents geotagged thermal images, anomaly localization overlays, and real-time alerts, allowing operators to quickly assess pipeline health and take timely corrective action. The flowchart representation of the working principle is illustrated in Fig. 1.

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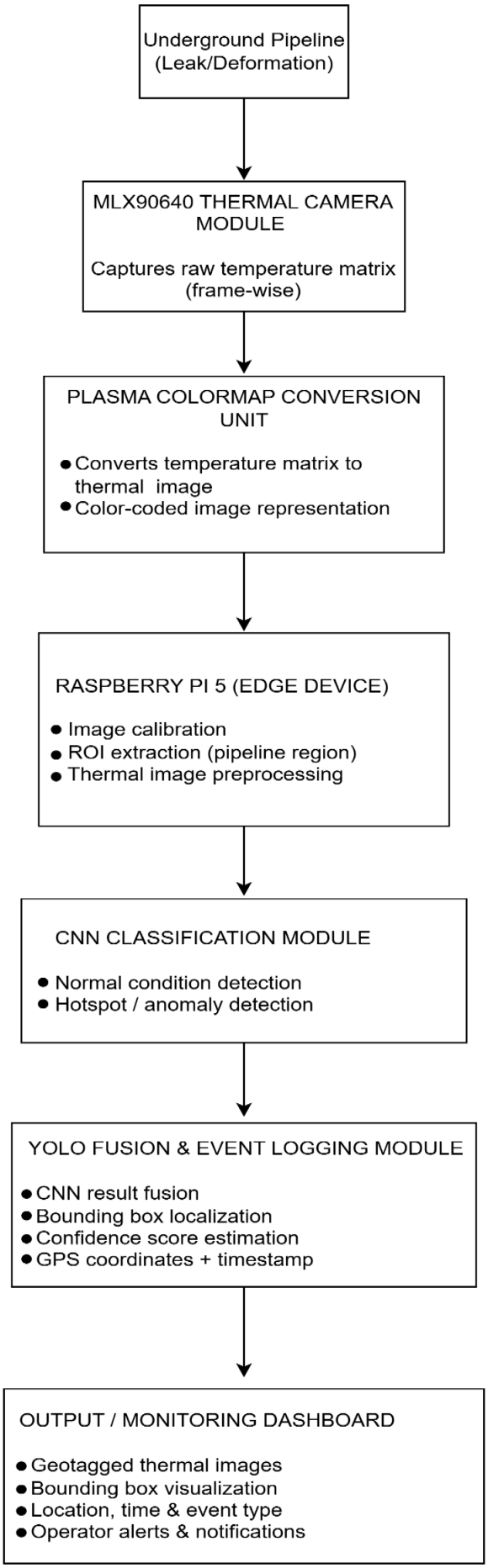


Figure 1. Flowchart of device workflow.

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Fig. 2 demonstrate the circuit connection for the step-up and the overall block diagram of the circuit is presented in Fig. 3. Table-I shows the components used in the proposed system.

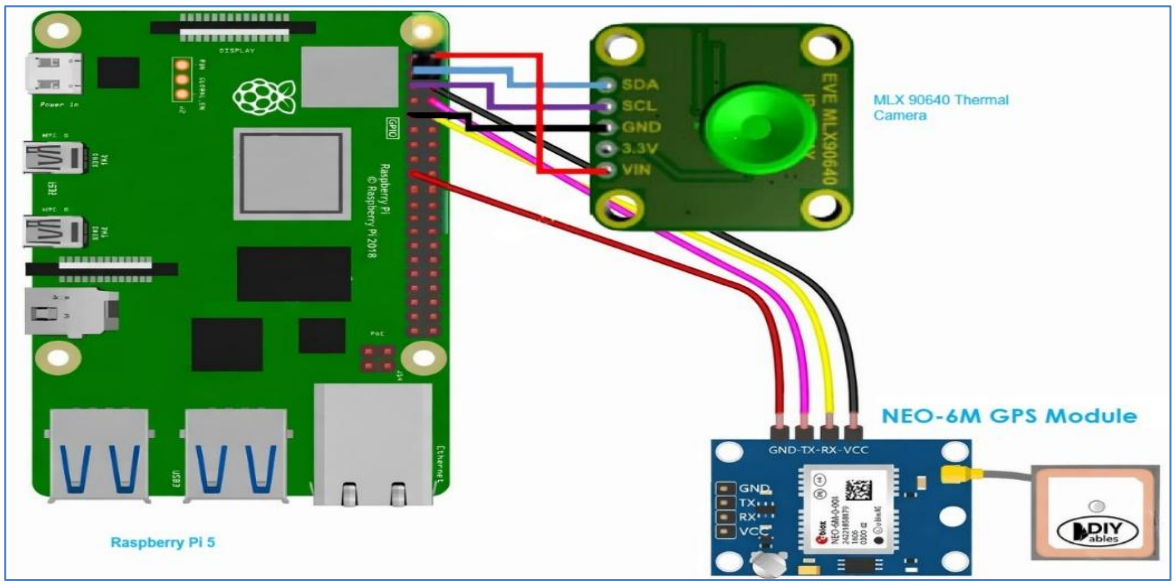


Figure 2. Circuit connections of the setup

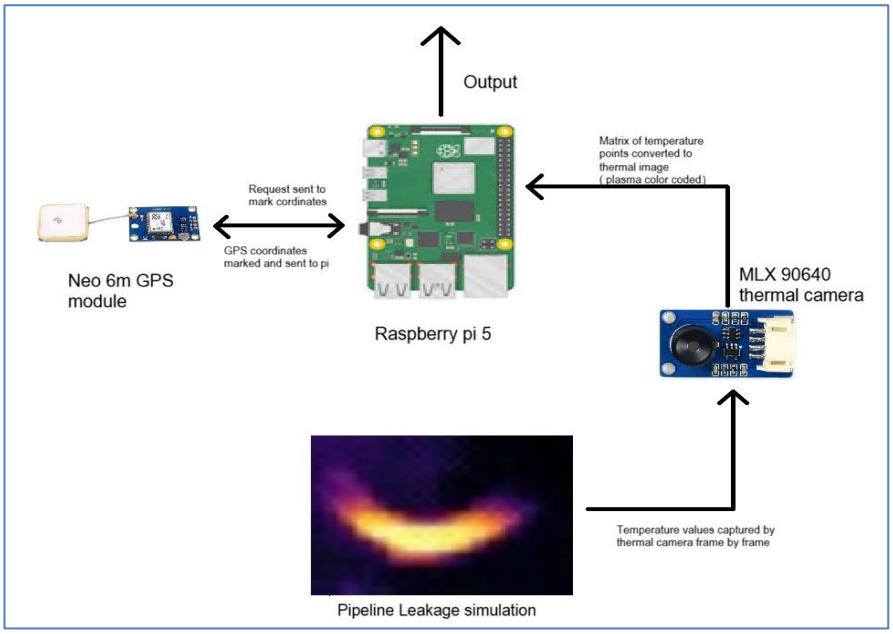


Figure 3. Block diagram of the proposed system.

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Table I: Hardware Component List

Name	Specification	Quantity	Purpose
Raspberry Pi 5	Quad-core 2.4 GHz Cortex A76 CPU, Video Core VII GPU, 16 GB RAM, 40 GPIO pins	1	Acts as the main processing unit for sensor input, camera control, and ML deployment
MLX 90640	32 * 24 p frame, 64hz I2C interface.	1	Captures thermal images and real time video stream.
Micro SD Card	Class 10, 32GB	1	Stores the OS, code, Python libraries, captured images and data.
Jumper Wires	Female-to-Female, standard size	8	Connect thermal camera and GPS module to Raspberry Pi GPIO pins.
9V 3A Power Supply/Power bank	27W USB-C Power	1	Provides stable power to the Raspberry Pi 5.
GPS(NEO-6M) module	5Hz position update rate. Configurable rates ranged from 4800 Baud to 115200 Baud rates. (default 9600) Supersense Indoor GPS: -162 dBm tracking sensitivity. Separated 18 x 18mm GPS antenna.	1	Location tracking and navigation

8. Experimental validation results:

A. System Idle Monitoring

Fig. 4 shows the setup of the Raspberry Pi 5, GPS(NEO-6M) module and in the Thermal camera (MLX 90640**)** idle (standby) state waiting for pipeline detection.

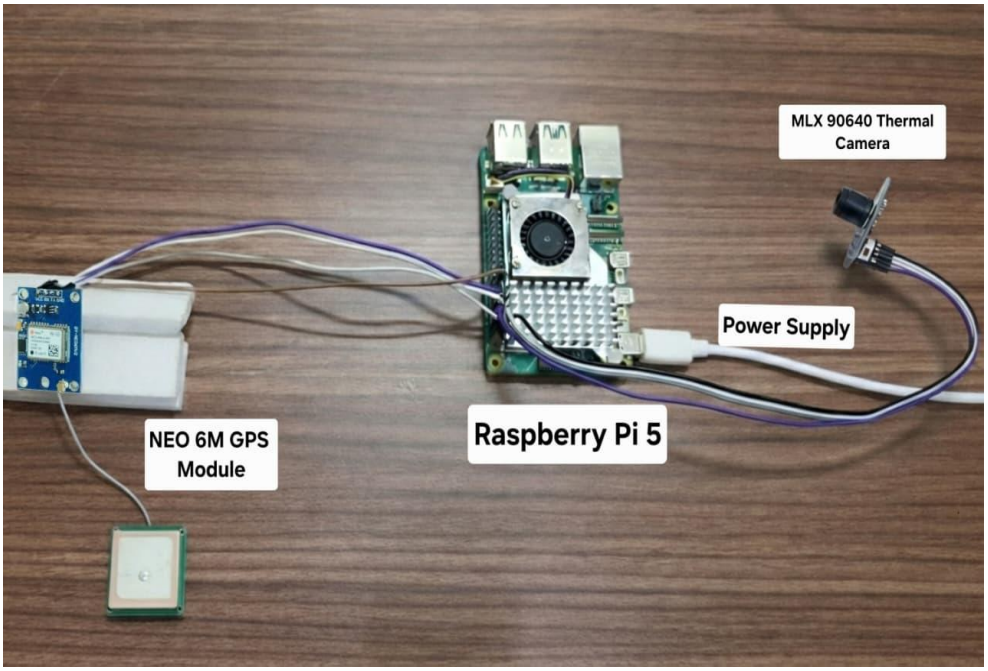


Figure 4. Initial state of the setup

B. Pipeline Heat Signature Detection via Thermal camera

The thermal camera senses infrared radiation from the pipeline, and the Raspberry Pi processes this data to highlight temperature anomalies, helping detect the pipeline path, leaks, or faults along with geolocation and timestamp (Fig. 5)

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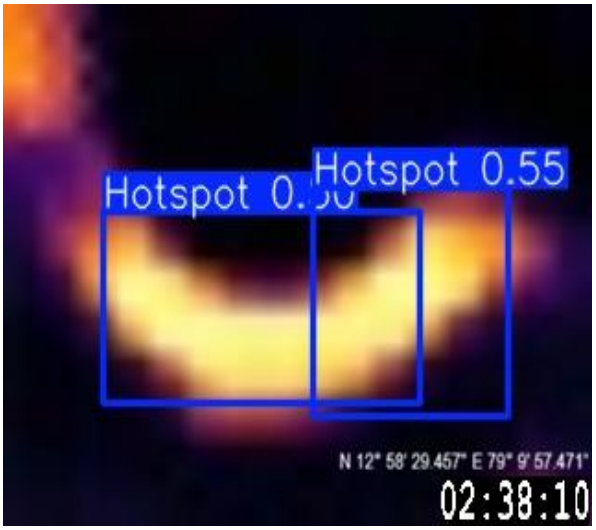


Figure 5. Image captured by the Thermal camera and modelled using YOLO v8s

C. Real-Time Image Capture

Fig. 6 illustrates the pixel by pixel captured image or video from the thermal camera.

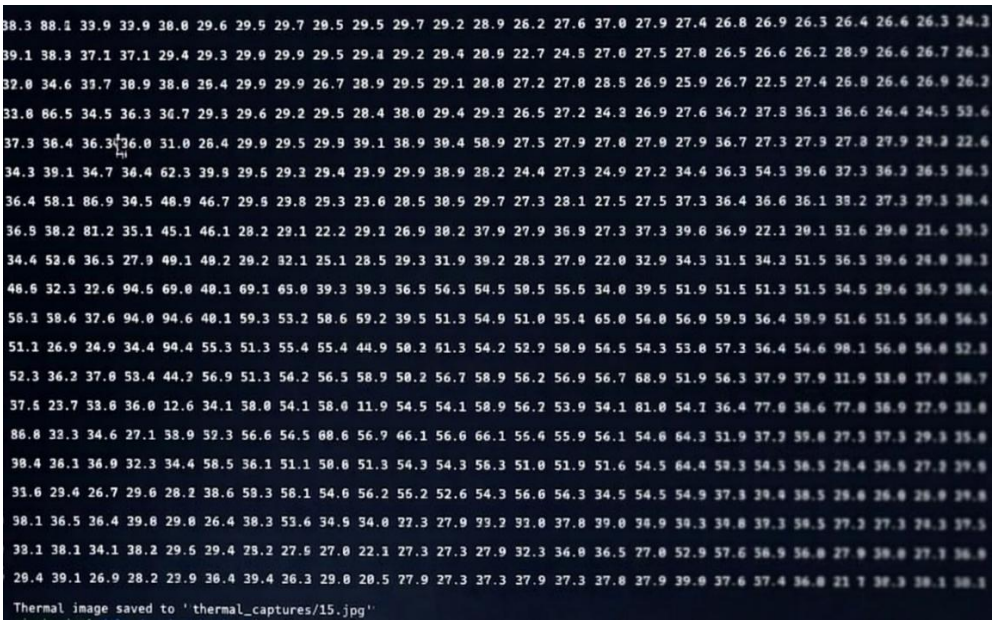


Figure 1. Console log of the python code to save the image file locally in SD Card.

D. YOLO v8s Image Processing

YOLOv8 processes thermal camera images to detect pipeline anomalies and validates detections using confidence scores, filtering out low-confidence results for accurate fault identification. This is shown in Fig. 7.

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```
Class: 0  Confidence: 0.559  Box: [184.12, 110.1, 319.76, 225.16]
Class: 0  Confidence: 0.478  Box: [46.32, 116.25, 317.76, 232.57]
Class: 0  Confidence: 0.386  Box: [0.0, 70.99, 187.37, 228.46]
Class: 0  Confidence: 0.21  Box: [0.0, 61.3, 54.13, 223.44]
Detection results for 51.jpg:
Class: 0  Confidence: 0.4  Box: [58.77, 126.8, 220.44, 232.49]
Class: 0  Confidence: 0.23  Box: [64.25, 140.78, 310.33, 237.17]
Class: 0  Confidence: 0.205  Box: [170.84, 171.64, 320.0, 240.0]
Detection results for 52.jpg:
Class: 0  Confidence: 0.674  Box: [0.85, 72.73, 300.79, 231.04]
Class: 0  Confidence: 0.388  Box: [0.0, 105.5, 269.24, 239.11]
Detection results for 53.jpg:
Class: 0  Confidence: 0.52  Box: [56.25, 126.82, 310.12, 236.24]
Class: 0  Confidence: 0.277  Box: [0.36, 159.91, 319.93, 240.0]
Class: 0  Confidence: 0.185  Box: [0.0, 172.25, 266.62, 240.0]
Detection results for 54.jpg:
Class: 0  Confidence: 0.332  Box: [258.36, 182.33, 320.0, 240.0]
Class: 0  Confidence: 0.29  Box: [0.0, 99.35, 148.08, 232.37]
Class: 0  Confidence: 0.208  Box: [9.63, 187.88, 221.99, 240.0]
Detection results for 55.jpg:
Class: 0  Confidence: 0.448  Box: [61.09, 113.12, 307.21, 231.4]
Class: 0  Confidence: 0.297  Box: [126.71, 143.36, 320.0, 236.16]
Class: 0  Confidence: 0.161  Box: [95.05, 156.18, 320.0, 240.0]
Detection results for 56.jpg:
Class: 0  Confidence: 0.531  Box: [178.77, 139.23, 320.0, 233.15]
Class: 0  Confidence: 0.481  Box: [0.0, 100.36, 59.19, 229.81]
Class: 0  Confidence: 0.208  Box: [48.13, 159.85, 317.78, 240.0]
Detection results for 57.jpg:
Class: 0  Confidence: 0.492  Box: [42.42, 143.39, 278.7, 233.04]
Class: 0  Confidence: 0.326  Box: [0.19, 166.24, 284.41, 240.0]
Detection results for 58.jpg:
Class: 0  Confidence: 0.548  Box: [0.0, 52.92, 312.42, 240.0]
Detection results for 59.jpg:
Class: 0  Confidence: 0.554  Box: [0.2, 98.92, 165.0, 203.53]
Detection results for 8.jpg:
Class: 0  Confidence: 0.285  Box: [150.8, 2.87, 224.98, 122.17]
Detection results for 9.jpg:
Processed 52 image(s). Outputs saved to: C:\Projects\Training_fine_tune\outputs
(Venv) PS C:\Projects\Training_fine_tune>
```

Figure 2. Thermal Image Analysis Output with Associated Detection Confidence and Spatial Coordinates.

E. Extracted Geolocation and Timestamp

For every detected pipeline anomaly, the system automatically extracts precise geolocation coordinates (latitude and longitude) from the onboard GPS module along with a synchronized timestamp. This metadata is tightly coupled with the corresponding thermal image frame and detection output, ensuring accurate spatial and temporal context for each event. The geotagged information enables exact localization of underground pipeline deformations on digital maps, supports traceability of inspection data, and allows operators to analyze when and where anomalies occur, facilitating timely maintenance actions and historical trend analysis (Fig. 8).

```
PS C:\Projects\Training_fine_tune> python -u "c:\Projects\Training_fine_tune\GPS.py"
27.jpg saved | GPS: N 12° 58' 29.457" E 79° 57' 47.41" | time: 02:37:50
28.jpg saved | GPS: N 12° 58' 29.457" E 79° 57' 47.41" | time: 02:37:51
29.jpg saved | GPS: N 12° 58' 29.457" E 79° 57' 47.41" | time: 02:37:52
30.jpg saved | GPS: N 12° 58' 29.457" E 79° 57' 47.41" | time: 02:37:53
31.jpg saved | GPS: N 12° 58' 29.457" E 79° 57' 47.41" | time: 02:37:54
32.jpg saved | GPS: N 12° 58' 29.457" E 79° 57' 47.41" | time: 02:37:55
33.jpg saved | GPS: N 12° 58' 29.457" E 79° 57' 47.41" | time: 02:37:56
34.jpg saved | GPS: N 12° 58' 29.457" E 79° 57' 47.41" | time: 02:37:57
35.jpg saved | GPS: N 12° 58' 29.457" E 79° 57' 47.41" | time: 02:37:58
36.jpg saved | GPS: N 12° 58' 29.457" E 79° 57' 47.41" | time: 02:37:59
37.jpg saved | GPS: N 12° 58' 29.457" E 79° 57' 47.41" | time: 02:38:00
38.jpg saved | GPS: N 12° 58' 29.457" E 79° 57' 47.41" | time: 02:38:01
39.jpg saved | GPS: N 12° 58' 29.457" E 79° 57' 47.41" | time: 02:38:02
40.jpg saved | GPS: N 12° 58' 29.457" E 79° 57' 47.41" | time: 02:38:03
41.jpg saved | GPS: N 12° 58' 29.457" E 79° 57' 47.41" | time: 02:38:04
42.jpg saved | GPS: N 12° 58' 29.457" E 79° 57' 47.41" | time: 02:38:05
43.jpg saved | GPS: N 12° 58' 29.457" E 79° 57' 47.41" | time: 02:38:06
44.jpg saved | GPS: N 12° 58' 29.457" E 79° 57' 47.41" | time: 02:38:07
45.jpg saved | GPS: N 12° 58' 29.457" E 79° 57' 47.41" | time: 02:38:08
46.jpg saved | GPS: N 12° 58' 29.457" E 79° 57' 47.41" | time: 02:38:09
PS C:\Projects\Training_fine_tune> & C:/Projects/Training_fine_tune/Venv/Scripts/Activate.ps1
(Venv) PS C:\Projects\Training_fine_tune>
```

Figure 8. Output Display of Synchronized Image Capture, GPS Coordinates, and Timestamp Information

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9. What aspect(s) of the invention need (s) protection?

The following elements require protection:

- The detection of pipeline deformation using thermal imaging along with the time stamp and Geolocation information.
- The integrated infrastructure of the layout of proposed system.
- The real-time AI based pipeline deformation detection using CNN.

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10. What is Technology readiness level of your invention? (Tick the appropriate TRL)

Research			Development			Deployment		
TRL 1	**TRL 2**	**TRL 3**	**TRL 4**	**TRL 5**	**TRL 6**	**TRL 7**	**TRL 8**	**TRL 9**
Basic Principles observed	Technology concept formulated	Experimental proof of concept	Technology validated in a lab	Technology validated in a relevant environment (industrially relevant in case of key enabling technologies)	Technology demonstrated in a relevant environment (industrially relevant in case of key enabling technologies)	System prototype demonstration in an operational environment	System complete and qualified	Actual system proven in an operational environment (competitive manufacturing in case of key enabling technologies, or in space)
			✓					

----------END OF THE DOCUMENT----------